

Skills

Preparing Injections from a Vial

Checklist

Checklist

S = Satisfactory U = Unsatisfactory NP = Not Performed

Step	S	U	NP	Comments
Verified the health care provider's orders.	☐	☐	☐	
Performed hand hygiene and prepared supplies.	☐	☐	☐	
Prepared medications:	☐	☐	☐	
1. When using an automated medication dispensing system, logged on and accessed the patient's profile. 2. Selected vial of the prescribed drug from the medication cart or automated dispensing system. Compared the name of the drug on the label of the vial with the name in the MAR. 3. Checked the expiration date and drug concentration or dose on the vial. Calculated the correct volume to withdraw. Double-checked calculation. Asked another RN to check calculation if needed. 4. Checked agency policy to note which high-alert medications require validation by a second RN.				
Prepared the vial containing a solution:	☐	☐	☐	
1. Removed the cap on an unused vial to expose the rubber seal. Firmly wiped the surface of the rubber seal with an alcohol swab. Allowed it to dry. 2. Selected a syringe, and attached a needle or needleless access device. If the syringe was capped, removed the cap first. Reviewed agency's policy or the drug insert to determine if it was necessary to use a filter needle. 3. Pulled back on the plunger to draw a volume of air into the syringe equivalent to the volume of medication to be aspirated from the vial. 4. With the vial on a flat surface, applied pressure to insert the tip of the needle or needleless device through the center of the rubber seal. 5. Injected the air drawn into the syringe into the air space of the vial, firmly holding the plunger. 6. Held the vial with nondominant hand. Held the syringe barrel and plunger with dominant hand. Inverted the vial securing the plunger. Kept the tip of the needle or needleless device below the level of the fluid. 7. Allowed the air pressure from the vial to fill the syringe gradually. If necessary, pulled back slightly on the plunger to hasten the process. 8. When the level of medication in the syringe reaches the desired volume, positioned the tip of the needle or needleless device into the air space of the vial. Tapped the syringe barrel				

gently to dislodge any air bubbles. If air remains at the top of the syringe, ejected any remaining air into the vial.

9. Rechecked the volume in the syringe.
 10. Removed the needle or needleless access device from the vial. Avoided pulling the plunger.
 11. Held the syringe vertically, needle up, at eye level, and ensured the correct volume and the absence of air bubbles. Removed the remaining air by tapping the barrel to dislodge the air bubbles. If air remains, drew back slightly on the plunger, and then pushed it upward to eject the air. Did not eject any fluid. Rechecked the volume of the drug in the syringe. Covered the needle with a safety sheath or cap.
 12. When injecting the medication into the patient's tissue, changed the needle to the appropriate gauge and length, according to the route of medication administration.
 13. When preparing the medication away from the patient's bedside, clearly labeled the syringe with the name of the medication and the dose in the syringe.
 14. For a new multi-dose vial, wrote the date on which it was opened and initials on the vial label. Removed the cap and swabbed the rubber seal with an alcohol swab. Let it dry.
 15. When a multi-dose vial was already open, wiped the surface of the rubber seal with an alcohol swab and let it dry.
 16. Discarded single-dose vials after use. Discarded multi-dose vials per the organization's practice.
 17. Had another nurse to verify the medication per the organization's practice.
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Prepared a vial containing a powder:



1. Removed the cap on the vial of powdered medication and the cap on a vial of the proper diluent. Made sure to check the manufacturer's guideline for the correct diluent and volume of diluent to use, and if use of a filter needle was recommended. Wiped both rubber seals with an alcohol swab, and allowed the alcohol to dry.
 2. Used a needle or needleless device to draw the appropriate type and volume of diluent suggested by the manufacturer into the syringe.
 3. Inserted the tip of the needle or the needleless device through the center of the rubber seal of the vial of powdered medication. Injected the measured diluent into the vial containing the powder. Withdrew and disposed of the syringe or access device in a puncture proof container.
 4. Rolled the vial between the palms of hands to mix the powder and diluents thoroughly. Did not shake the vial.
 5. Read the label carefully to determine the concentration of the drug.
 6. Wiped off the top with an alcohol swab and allowed it to dry. Drew up the appropriate volume of medication based on its concentration in the reconstituted formulation into a syringe.
 7. For medications in vials that contain both the medication and diluents separated by a gasket:
 1. Pressed down on the plastic activator to force the diluent into the lower compartment.
 2. Removed the top and gently agitated the vial to mix the solution.
 3. Swabbed the gasket with alcohol. Inserted the needle through the center of the stopper.
 4. Inverted the vial and withdrew the dose.
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Asked another nurse to verify the medication per the organization's practice.



When drawing up medication away from the patient’s bedside, labeled the syringe with the name of the medication and the amount of medication or dose in the syringe.

Disposed of used supplies. Discarded multi-dose vials per the organization’s practice. Placed all single-dose and empty multidose vials, and the needle or needleless access device in the appropriate disposal container. Cleaned work area, and performed hand hygiene.

Documented the procedure in the patient’s record.

Learner:

Evaluator:

Date:

Signature:

Signature: